

AGENDA:

- Prime NO.
- Generate all prime no. from L to A
- Count of divisors
- Pair with given sum
- Count pair sum K
- Pair diff K

Rules

- Answer to Private chat
- Question to Question Tab
- 100%



Prime Numbers

→ A natural no. with exactly 2 factors

1 is not a prime no.

< Question > : Given a number N. Check if it is prime or not.

[Approach] :

```
int countDiv ( int n ) {  
    count = 0  
  
    for ( i = 1 ; i*i <= n ; i++ ) {  
        if ( n%i != 0 ) continue;  
        // i & n/i are factors  
        if ( i*i == n ) { count++ }  
        else { count += 2 }  
    }  
    return count  
}
```

TC: $O(\sqrt{N})$

Check Prime

```
if countDiv(n) == 2    ✓ Prime  
else                  X not prime
```

Real World Example

SecurePrime Inc. a renowned cybersecurity firm, is on a mission to upgrade its **encryption** techniques to outsmart potential attackers. Their strategy involves utilizing **prime numbers**, known for their pivotal role in strengthening cryptographic systems

Problem

SecurePrime Inc. plans to enhance its **encryption keys** by incorporating **random prime numbers** from the **1 to A** into their algorithms. This approach significantly complicates any brute force attempts by attackers, making the encryption much more robust and reliable.

Task

As the requirement , you need to generate prime numbers** from the **1 to A**

Input A

Output All primes in 1 to A

Eg A = 10

O/p 2 3 5 7



< Question > : Given an integer N. Check every number from 1 to N if it is a prime number or not.

N = 10 [1 2 3 4 5 6 7 8 9 10]

$1 \leq N \leq 10^6$

ans[] $\rightarrow$ [X T T F T F T F F F]

Brute force

for val $\rightarrow$ 1 to N

check if val is prime or not

TC: $O(N * \sqrt{N})$

$N = 10^6$

$10^6 * 10^3 = \sim 10^9$ iterations



$\therefore < 10^8$ iterations are fine



Sieve of Eratosthenes



Idea -2

Dipika { CT }

Mhd

4 6 8 50

Anil

6 9 12 48

Ravi

5*5 5*6 5*10

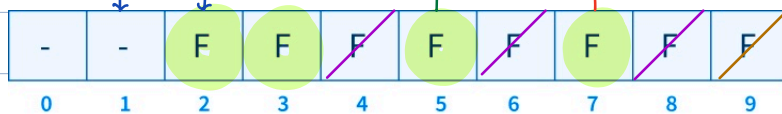
harshad 7*7 7*8

Priyank { monitor }

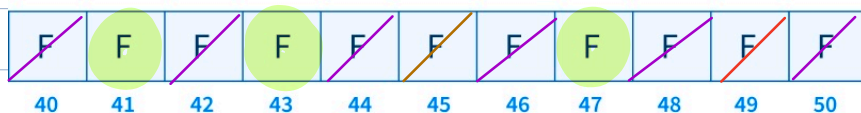
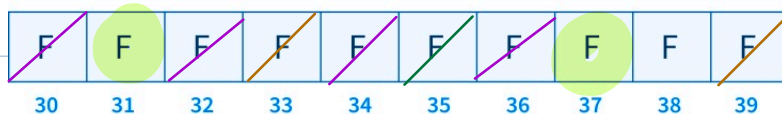
MHD

havi

harshad



Anil





Pseudocode

```
isPrime[n+1] // boolean
```

```
∀i isPrime[i] = true
```

```
isPrime[0] = false
```

```
isPrime[1] = false
```

```
for (int i=2 ; i*i <= n ; i++) {
```

```
    if (isPrime[i]) {
```

```
        // mark multiples as false
```

```
        for (j = i*i ; j <= N ; j+=i ) {
```

```
            isPrime[j] = false
```

```
        }
```

```
    }
```

```
}
```

TC: $O(N \log(\log(n)))$

SC: $O(N)$

F T	F T	T	T	F T	T	F T	T	F T	F T	F T
0	1	2	3	4	5	6	7	8	9	10

**T.C Analysis**

N	~ iterative	
2	4 6 8 10 N	$\sim N/2$
3	9 12 N	$\sim N/3$
4	X	
5	25 30 N	$\sim N/5$
6	X	
⋮		
$\sqrt{N}$		

$$TC \quad \frac{N}{2} + \frac{N}{3} + \frac{N}{5} + \frac{N}{7} + \frac{N}{11} + \dots$$

$$N \left\{ \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \dots \right\}$$

sum of reciprocal of primes

 $\log(\log(n))$ $\log n$

$$\left\{ \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \dots \right\} \leq \left\{ 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots \right\}$$

sum of reciprocal of
all natural no.

$$\int_1^N \frac{1}{x} dx \rightarrow \log x$$

Q> Given N Count all divisors from 1 to N

q/p

N = 10

	1	2	3	4	5	6	7	8	9	10
Divisors	1	2	2	3	2	4	2	4	3	4

o/p

Bruteforce $\rightarrow$ for val $\rightarrow$ 1 to N
count Div(val) $\rightarrow O(\sqrt{N})$ } TC: $O(N\sqrt{N})$

Approach 2

Sieve

	0	1	2	3	4	5	6	7	8	9	10
factors	0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
		1	1	1	1	1	1	1	1	1	1
			2	2	2	2	2	2	2	2	2
					3		3		3	3	3
							4		4		4
	0	1	2	2	3	2	4	2	4	3	4

Pseudocode

factors [N+1]

$\forall i \text{ factors}[i] = 0$

```
for (i = 1 ; i <= N ; i++) {  
    |  
    for (j = i ; j <= N ; j += i) {  
        |  
        factors[j] += 1  
    }  
}
```

TC: $O(N \log N)$

TC

N	~ iterations
1	N
2	$N/2$
3	$N/3$
4	$N/4$
	$\vdots$

TC $N + \frac{N}{2} + \frac{N}{3} + \frac{N}{4} + \dots$

$N \{ 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \}$

sum of reciprocal of natural no. $\log N$

Break 22 : 27



< Question > : Given an integer sorted array A and an integer K. Find any pair [i, j] such that $A[i] + A[j] = K$, $i \neq j$

Example : $A[] \rightarrow [-5, -2, 1, 8, 10, 12, 15],$

0 1 2 3 4 5 6

ans true

K = 11

Quiz : $[-3, 0, 1, 3, 6, 8, 11, 14, 18, 25],$

0 1 2 3 4 5 6 7 8 9

ans true

K = 12

**BF Approach**TC: $O(N^2)$

```

for i  $\rightarrow$  0 to N-1 {
    for j  $\rightarrow$  i+1 to N-1 {
        if ( $A_i + A_j == k$ ) return true
    }
}
return false

```

Binary Search

**Idea -2** $a + b == k$ $\uparrow$

know

 $b = k - a$



for $i \rightarrow 0$ to $N-1$

TC: $O(N \log N)$

$a = A[i]$

$b = k - a$

// find b using binary search

$i+1$ $N-1$

$A[] \rightarrow [-5, -2, 1, 8, 10, 12, 15],$

$K = 11$

0 1 2 3 4 5 6

Idea 3 **set**

$a + b = k$

TC: $O(N)$

$b = k - a$

SC: $O(N)$

hmap = {}

for $i \rightarrow 0$ to $N-1$ {

$a = A[i]$

$b = k - a$

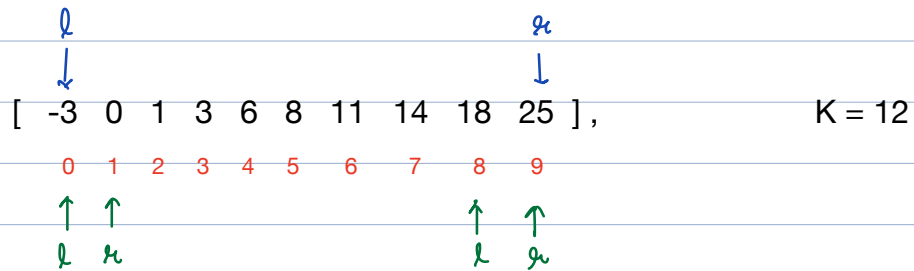
// 0..... $i-1$

set

if b exists in set { return true }

}

Idea -4 Two - pointer approach



1. $l = 0, r = 1$

2. $r = N-1, l = N-2$

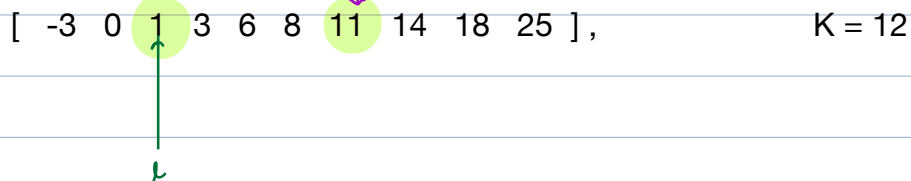
Not helpful

3. $l = 0, r = N-1$

$A = [-3, 0, 1, 3, 6, 8, 11, 14, 18, 25]$, $K = 12$

l

$$A_l + A_r = 15$$



A_l	A_r	$A_l + A_r$	Decision
-3	25	22	$r--$
-3	18	15	$r--$
-3	14	11	$l++$
0	14	14	$r--$
0	11	11	$l++$
1	11	12	ans



</> Code

 $l = 0$ $r = N - 1$

```
while ( l < r ) {  
    total = A[l] + A[r]  
    if ( total == k ) {  
        return true  
    }  
    else if ( total < k ) {  
        l++  
    } else {  
        r--  
    }  
}
```

TC: $O(N)$ SC: $O(1)$

return false

l r

[-3 0 1 3 6 8 11 14 18 25],

 $K = 12$



< Question > : Find count of all the pairs in a sorted array whose sum is K. ($i \neq j$)

arr[] $\rightarrow$ [1 3 4 5 6 7 10], arr = 2
0 1 2 3 4 5 6 K = 10

arr[] $\rightarrow$ [1 3 4 4 5 6 7 10] arr = 3
0 1 2 3 4 5 6 7

arr[] $\rightarrow$ [1 3 4 4 5 6 6 6 7 10] arr = 7
0 1 2 3 4 5 6 7 8 9

BF $\rightarrow$ for all pairs check $A_i + A_j == K$ TC: $O(N^2)$
arr[] $\rightarrow$ [1 3 4 4 5 5 5 5 5 6 6 6 7 10]

Binary search ? TC: $N \log N$

arr[] $\rightarrow$ [1 3 4 4 5 6 6 6 7 10] K = 10
0 1 2 3 4 5 6 7 8 9

need = 6

For all val in AT, need = K - val find the freq of need
add freq of need to ans

	1	3	4	4	5	5	5	6	6	6	7	10
	↓	↓	↓	↓	↓	↓	↓	↓				
need	9	7	6	6	5	5	5	4				
ans		+1	+3	+3	+2	+1	+0	0				

idea 3 > use hashmap and keep track of freq from range (0..... i-1)

	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	$k=10$
Arr	1	3	4	4	5	5	5	6	6	6	7	10	
need	9	7	6	6	5	5	5	4	4	4	3	0	
ans	+0	+0	+0	+0	+0	1	+2	+2	+2	+2	+1	0	

HMAP {	1 : 1	$T.C: O(N)$
	3 : 1	$S.C: O(N)$
	4 : 2	
	5 : 3	
	6 : 3	
	7 : 1	
	10 : 1	
	}	



Only distinct elements

arr[] $\rightarrow$ [1 3 4 5 6 7 10], $K = 10$

$l = 0$

$r = N - 1$

$ans = 0$

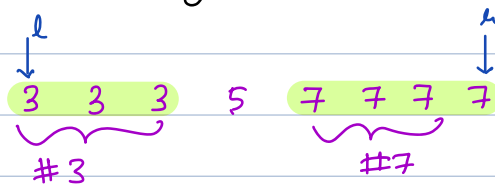
```
while ( l < r ) {  
    total = A[l] + A[r]  
    if ( total == k ) {  
        ans++ l++ r--  
    }  
    else if ( total < k ) {  
        l++  
    } else {  
        r--  
    }  
}
```

TC: $O(N)$

SC: $O(1)$

return false

Elements are repeating



$k = 10$

$$A_l + A_r = k$$

$$ans += \#3 * \#7$$

$l = 0$

$r = N - 1$

$ans = 0$

TC: $O(N)$

SC: $O(1)$

while ($l < r$) {

total = $A[l] + A[r]$

if (total < k) {

l++

} else if {

r--

}

else { // total == k

// find the freq of $A[l]$

freql

// find the freq of $A[r]$

freqr

if ($A[l] == A[r]$) {

ans += $(freql * (freql - 1)) / 2$

} else

ans += $freql * freqr$

while ($l \leq r$ && $A[l] == A[r]$)

freql++

l++

}

$A[] = [1, 5, 5, 5, 6]$ $k = 10$

freql = 3
freqr = 3

3

3

3

$$ATT = \begin{array}{c} \downarrow \quad \frown \quad \frown \quad \frown \quad \downarrow \\ 2 \quad 2 \quad 2 \quad 2 \quad 2 \end{array}$$

$$k = 4$$

$$\text{freq } l = 5$$

$$\text{freq } k = 5$$

$$\frac{5 * 4}{2} = \underline{\underline{10}}$$



$$K \geq 0$$

< Question > : Given an integer sorted array A and an integer K. Find any pair $[i, j]$ such that $A[i] - A[j] = K$, $i \neq j$

$$K = 11$$

arr[] $\rightarrow$ [-5 -2 1 8 10 12 15]

0 1 2 3 4 5 6



Idea -1

BF same as above



Idea -2

BS same as above



Idea -3

HMap same as above



Idea -4

Two pointers



Two Pointers

SCALER

arr[] → [-5 -2 1 8 10 12 15] k = 11

l ↓ l ↓ r ↓ r ↓
 0 1 2 3 4 5 6
 ↑ ↑

A_l	A_r	$A_r - A_l$	Decision
-5	15	20	multiple options
-2	15	17 ↓	l++
-5	12	17 ↓	r--

l r

arr[] → [-5 -2 1 8 10 12 15] k = 11

0 1 2 3 4 5 6
 ↑ ↑

A_l	A_r	$A_r - A_l$	Decision
-5	-2	3	$r++$
-5	1	6	$r++$
-5	8	13	$r++$
-2	8	10	$r++$
-2	10	12	$r++$
1	10	9	$r++$
1	12	11	ans

HW



Doubt series

-	-	F	F	F	F	F	F	F	F
0	1	2	3	4	5	6	7	8	9

F	F	F	F	F	F	F	F	F	F
10	11	12	13	14	15	16	17	18	19

F	F	F	F	F	F	F	F	F	F
20	21	22	23	24	25	26	27	28	29

F	F	F	F	F	F	F	F	F	F
30	31	32	33	34	35	36	37	38	39

F	F	F	F	F	F	F	F	F	F	F
40	41	42	43	44	45	46	47	48	49	50

set

val = 5

for i in set :

5 * i

